

Cover Page	
Project Title	<i>Application of Spread Option in LNG Geographical Arbitrage</i>
Name	<i>Li Shuai</i>
Student ID	<i>A0130504Y</i>
Itemized workload	
Data collection	<i>25 hours</i>
Literature review	<i>35 hours</i>
Programming	<i>45 hours</i>
Framework Setup	<i>45 hours</i>
Total workload	<i>150 hours</i>
Original contribution	
<i>This project has the following original contributions:</i> • Conceptualization of Physical LNG Trading: Developed a structured framework for analyzing LNG trading strategies • Quantification of Geographical Arbitrage: Established a methodology to measure and optimize physical cargo arbitrage opportunities • Spread Option Derivation: Derived a pricing model based on spread options framework tailored for LNG physical trading • Improved Pricing Models: Integrated real-time market feeds and stress-testing techniques to improve pricing models. • Scenario Analysis for Risk Assessment: Evaluated hedging strategy resilience under extreme market conditions via scenario analysis.	

APPLICATION OF SPREAD OPTION IN LNG GEOGRAPHICAL ARBITRAGE FE110 Project

Li Shuai A0130504Y

March 2025

Abstract

This study examines the application of spread options in LNG physical trading, focusing on their role in optimizing contractual flexibility and enhancing the monetization of geographical arbitrage opportunities. By integrating spread option valuation into physical cargo economics, we aim to provide a structured framework for managing embedded optionality in LNG contracts. Utilizing financial contract data from the Intercontinental Exchange (ICE), cargo flow analytics from Kpler, and various other market data sources, we conduct a comprehensive analysis of LNG shipping economics and apply a spread option valuation framework to assess its impact on trading strategies. Our findings indicate that proactively managing embedded optionality in LNG cargo contracts can significantly reduce financial risk exposure and improve profit and loss (PnL) stability across different market conditions. Moreover, leveraging spread option techniques enables companies to enhance the valuation of long-term contractual flexibilities, ensuring more effective risk-adjusted decision-making. This research contributes to the LNG industry by providing an analytical foundation for integrating financial derivatives into physical trading strategies. Future research can expand on these insights by exploring the valuation of additional types of optionality embedded in long-term LNG supply agreements, such as destination, volume, and pricing flexibility.

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1 Introduction

1.1 Objectives

The liquefied natural gas (LNG) market plays an increasingly critical role in the global energy landscape, making it essential for consumers, producers, investors, and industry stakeholders to understand its complexities. Given the growing demand for LNG as a transitional fuel in the shift toward cleaner energy sources, optimizing trading strategies and risk management practices is imperative. This study introduces a structured framework to evaluate physical arbitrage opportunities in LNG trading, enabling market participants to make informed decisions, mitigate risks, and maximize value in this dynamic sector.

This research systematically examines the mechanics of LNG physical cargo trading, with a particular emphasis on shipping economics under various market scenarios. Specifically, the study focuses on cargo diversion and optimization strategies, which are key drivers of profit and loss (PnL) in physical LNG trading. Geographical arbitrage, the practice of redirecting cargoes to higher-value markets, is central to this analysis. Beyond traditional physical trading frameworks, this study incorporates financial modeling techniques, including Stochastic Differential Equations, Black-Scholes-Merton model, Change of Numeraire method and Kirk's approximation, to integrate spread option valuation into LNG trading strategies. By applying these quantitative tools, traders, producers, and consumers can make data-driven decisions to efficiently monetize arbitrage opportunities through strategic cargo redirection.



Figure 1.1: LNG Export & Import Shipping Routes [1]

1.2 LNG Market and Geographical Arbitrage

The global LNG market is undergoing rapid expansion, driven by increasing demand in the Asia-Pacific and European regions, rising supply from major producers such as the United States, Qatar, and Australia, and substantial investments in infrastructure and technology. As nations accelerate their transition toward cleaner energy sources, LNG is positioned as a crucial component due to its lower carbon footprint compared to coal and oil [2]. However, the market remains highly volatile, influenced by geopolitical tensions, regula-

tory developments, environmental concerns, and fluctuating prices. In this dynamic environment, effective monitoring of geographical arbitrage [3] opportunities is essential for traders and market participants to proactively adjust strategies, mitigate risks, and maximize returns from their LNG portfolios.

With the increasing scale and complexity of LNG portfolios, market participants are diversifying their supply sources to enhance flexibility and hedge against regional price fluctuations. Major industry players are securing offtake agreements and equity positions across multiple basins to optimize cargo allocation. For instance, Woodside Energy Group Ltd, a leading LNG producer in the Pacific region, has significantly expanded its portfolio by securing long-term agreements with U.S.-based suppliers, including Mexico Pacific Limited [4] and Tellurian [5], adding an additional 1.3 million and 8 million tons per annum (MTPA) of LNG supply, respectively. This expansion, alongside Woodside’s equity holdings in the North West Shelf (NWS), Wheatstone, and Pluto gas fields, provides the company with enhanced optionality in cargo scheduling, allowing for strategic redirection to maximize value. As decision-making increasingly involves multi-continent destinations, the role of geographical arbitrage becomes more pronounced.

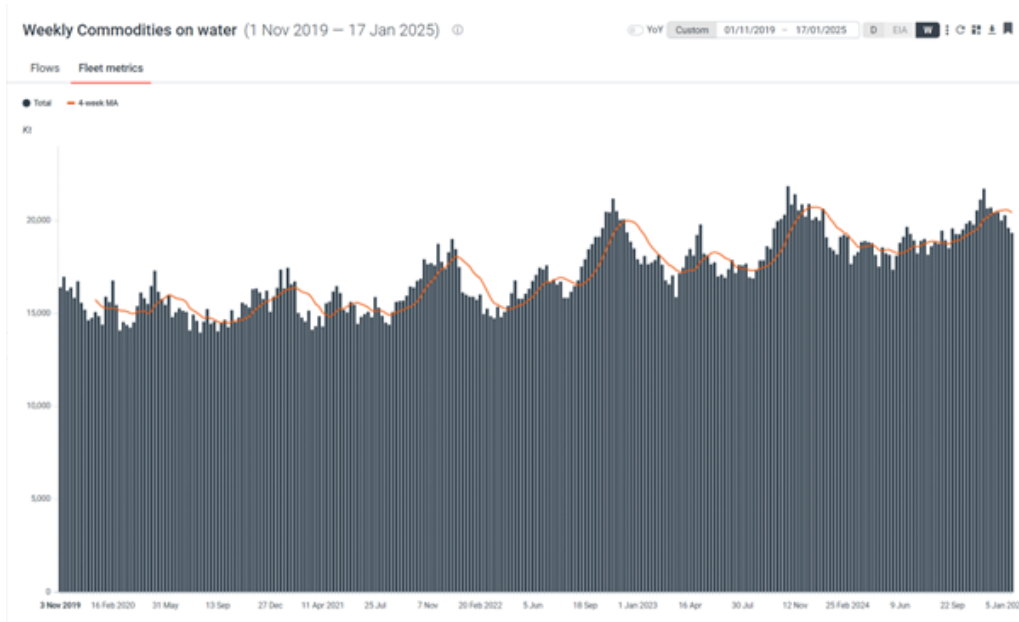


Figure 1.2: Weekly LNG on Water Increases Over Time [6]

Geographical arbitrage, which exploits price differentials across regional LNG markets such as Asia and Europe, enables traders to capitalize on temporary pricing imbalances. These disparities are driven by factors such as seasonal demand fluctuations, geopolitical disruptions, and localized supply constraints. Arbitrage exists when the price at an alternative destination sufficiently exceeds the base-case delivery price after accounting for incremental shipping costs. However, such opportunities are typically short-lived and infrequent due to the illiquid nature of physical LNG trading. A viable method to capture these arbitrage opportunities is through the use of spread options. By synthetically constructing a spread option that mirrors the embedded optionality in physical LNG contracts, traders can actively hedge exposure as the spread option moves in-the-money. Through delta hedging strategies, traders can systematically capture arbitrage value while dynamically adjusting positions based on market conditions. A detailed discussion of this approach will be further elaborated in Section 2.1.

The ability to efficiently monitor and execute on these arbitrage opportunities not only enhances revenue generation but also strengthens risk management and strategic planning. By leveraging advanced data analytics, real-time market intelligence, and sophisticated financial modeling techniques, stakeholders can optimize trade routes, improve asset utilization, and ensure more efficient resource allocation in an increasingly competitive global LNG market.

1.3 Underlying Financial Products

The LNG physical trading market utilizes a range of financial products representing different continents and regions worldwide. This study focuses on two primary underlying contracts: the Japan Korea Marker (JKM) [7] and the Title Transfer Facility (TTF) [8], which serve as the most widely used and representative gas indices in the LNG industry. The price spread between JKM and TTF futures, commonly referred to as the geographical spread, plays a critical role in LNG trading. JKM represents LNG delivery prices in Japan, Korea, and China, while TTF reflects delivery prices in Northwest European markets, such as France and Germany. Historically, these indices have exhibited strong correlations; however, disruptions caused by fundamental shifts in supply-demand balances, geopolitical events, and unforeseen market shocks occasionally lead to significant decoupling.

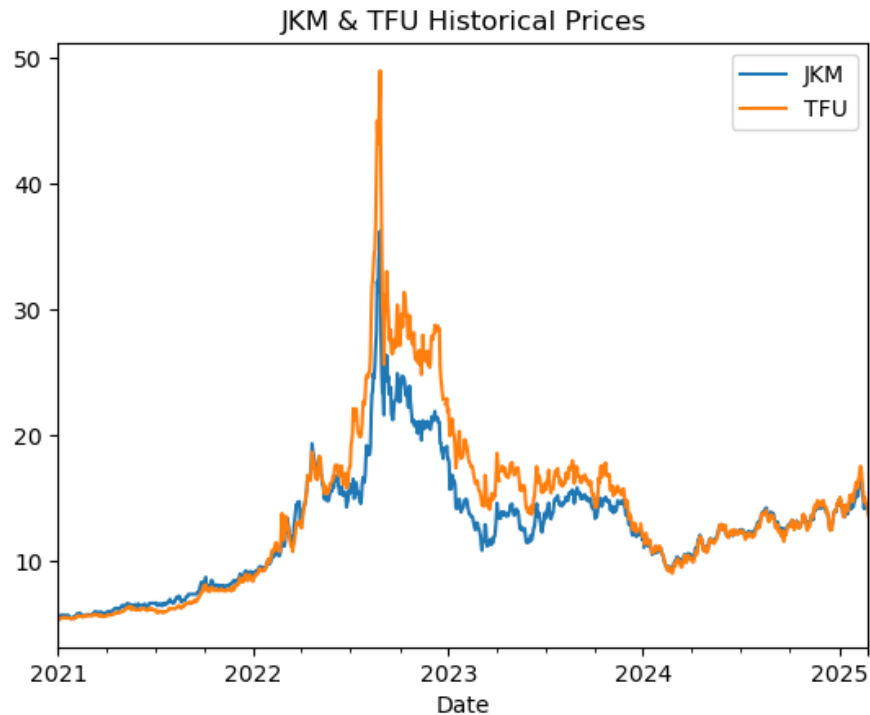


Figure 1.3: JKM & TTF Historical Prices

Both JKM and TTF futures are actively traded on the Intercontinental Exchange (ICE), with futures contracts serving as key instruments for hedging and speculation in the LNG market. The contract specifications for JKM and TTF futures can be found in their respective references [7] [8]. In this study, futures price data are directly sourced from the ICE API [9] to ensure accuracy and consistency in the analysis.

The LNG market is characterized by extreme price volatility, with annualized volatility reaching approximately 156% since January 2022. This heightened volatility is primarily driven by a combination of macroeconomic conditions, geopolitical risks, shifting supply-demand fundamentals, weather variations, and government policies. In general, prompt-month futures contracts with near-term expiration are more sensitive to geopolitical developments and seasonal demand fluctuations, while longer-dated contracts along the forward curve exhibit greater stability.

To accurately price and evaluate LNG spread options, additional market indices beyond JKM and TTF are required. These indices, along with historical price data, are extracted directly from ICE databases for comprehensive analysis.



Figure 1.4: JKM & TTF Historical Returns

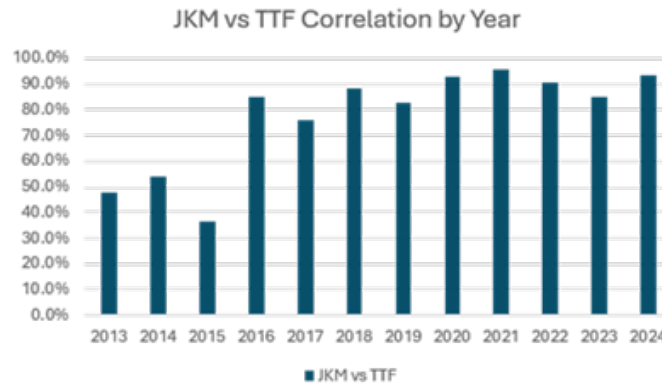


Figure 1.5: JKM & TTF correlation by Year

- **EUR/USD exchange rate:** Essential for converting transaction values between different currency bases.
- **Henry Hub index:** The U.S. benchmark for natural gas pricing, used as a reference for global pricing dynamics.

Apart from financial data, operational costs associated with LNG physical shipping must also be considered. Key components include:

- **Shipping costs and vessel operational specifications:** Critical for determining transportation expenses in LNG cargo valuation.

- **Port fees, shipping distances, and canal tolls:** Necessary for assessing the economic feasibility of arbitrage opportunities.

Shipping-related data are sourced from the Kpler database [6], while vessel-specific bunker consumption and structural information is obtained from Form B certificates. It is a regulatory safety certificate issued for cargo ships exceeding 500 gross tonnage (GT), ensuring compliance with international structural integrity, fire safety, and watertight regulations. Form B contains essential vessel operational information, such as Boil-off rate, Bunker consumptions, and etc. These operational factors collectively contribute to the assessment of physical cargo trading economics and spread option valuation in LNG trading.

FORM B

(a)	Ships Name	TBD
(b)	Builder and Yard	Daewoo Shipbuilding & Marine Engineering Co., Ltd
(c)	Hull No.	TBD
(d)	Year Built	2003
(e)	Port of Registry and Flag	TBD
(f)	IMO Number	TBD
(g)	Call Sign	TBD
(h)	Classification Society	DNV GL
(i)	Protection and Indemnity Club	Gard

1. Principal Particulars

(a)	Length overall	277 m
(b)	Length Between Perpendiculars	266 m
(c)	Breadth moulded	43.4 m
(d)	Depth moulded	26.0 m
(e)	Draught at summer freeboard (Extreme)	12.12 m
(f)	Height overall — keel to highest fixed point	52.95 m (49.50 mast down)
(g)	Maximum air draught (with full ballast and half bunkers) (corresponding draughts)	43.35 m at F 9.60m, A 9.60 (39.90 mast down)
(h)	Gross Tonnage (International)	93844 tons
(i)	Net Tonnage (International)	28154 tons
(j)	Gross Tonnage (Suez)	96870.92 tons
(k)	Net Tonnage (Suez)	83241.03 tons
(l)	Light Ship Displacement	29472 tons
(m)	Displacement (maximum)	77409.5 tones
(n)	Windage: Lateral (above deck)	608 m2
	Longitudinal (above deck)	2147 m2
	Longitudinal	6755 m2 at 9.42 m draft
	Longitudinal	5954 m2 at 12.37 m draft
(o)	Classification designation	+ 1A1 Tanker for Liquefied Gas, E0, NAUT-OC, LCS-SID, PLUS-1, NAUTICUS (Newbuilding)
(p)	Conditions of Carriage (as defined on Certificate of Fitness):	Liquefied Gases in Bulk

Figure 1.6: Form B Sample [10]

1.4 Physical Contracts and Spread Option

Physical commodity contracts often contain a broad range of embedded optionality, granting flexibility to both buyers and sellers in LNG trading. These embedded options significantly impact contract valuations and trading strategies. For instance, sellers typically retain the right to nominate delivery quantities within a predefined contractual range, usually 30 days prior to the cargo delivery window [11]. This contractual provision is mathematically equivalent to holding a long put option on the additional volume, with an expiry

date 30 days before the scheduled delivery. Such optionality provides sellers with downside protection while allowing them to optimize deliveries based on market conditions. Similarly, buyers may have the flexibility to nominate a discharge date within a pre-agreed delivery window. This enables them to maximize the value of their cargo by reselling it at a higher price when market conditions are favorable. Some more complex contract structures even include clauses allowing buyers to select an index from a predetermined list of commodity benchmarks, enabling them to price their cargo in the most advantageous way.

Including these flexibilities can be useful in contract negotiation as there is no common rules or framework to evaluate such terms. Counterparts may have different perceptions of optionality’s intrinsic and extrinsic value. Intrinsic value can be calculated directly based on observable market prices, whereas extrinsic value is subjective and relies on various valuation methodologies and assumptions. As market participants become more sophisticated in their approach to pricing and risk management, contractual flexibility continues to expand, facilitating smoother negotiations and improving overall market efficiency.

A spread option is a type of financial derivative where the payoff depends on the price difference between two underlying assets, commonly employed in commodities, energy markets, and financial trading to hedge against price differentials. In the context of LNG physical cargo trading, spread options play a crucial role in managing geographical arbitrage opportunities arising from price discrepancies between regional natural gas benchmarks, such as JKM and TTF. By synthetically replicating the optionality embedded in physical LNG contracts, spread options enable traders to hedge market exposure and monetize price differentials more effectively. This valuation framework quantifies the economic feasibility of cargo diversions by assessing the financial viability of re-routing shipments under varying market conditions. Additionally, it facilitates dynamic risk management, allowing traders to optimize portfolio decisions, capitalize on short-term arbitrage opportunities, and enhance overall operational flexibility within LNG trading strategies.

2 Methodologies

Closed form mathematical proof is conducted to ensure accuracy, and at the same time exposes critical assumptions worth paying attention to. The model starts with analyzing physical market followed by integrating spread option frameworks to further analyze physical trading opportunities.

Physical-cargo model analyzes cargo economics including purchase and sales contract prices at different indexes in different continental destinations, shipping costs, canal fees, port fees, and EUA costs. By comparing shipping economics at different destinations, we back calculated breakeven index spread prices which will be used as strike in the spread option model. Generic cargo size and shipping rotations are assumed for simplicity reasons. When opportunities arise in reality, assumptions would be updated accordingly.

The spread-option model assumes that the underlying assets follow Geometric Brownian Motion (GBM) with constant drift and volatility. The pricing is based on a European-style payoff, path-independence, and risk-neutral valuation under the assumption of arbitrage-free with a constant risk-free rate. The dynamics of the correlated Wiener processes are modeled using independent processes and Girsanov’s theorem to ensure the existence of a risk-neutral measure, satisfying Novikov’s condition for validity. A change of numeraire technique is employed by scaling one asset price by the other, under the assumption that the resulting ratio is lognormal. Finally, the spread option’s payoff is evaluated with a Black-Scholes-like formula. These assumptions collectively simplify the modeling process, and they could be further fine-tuned to follow real-world conventions with data from reality.

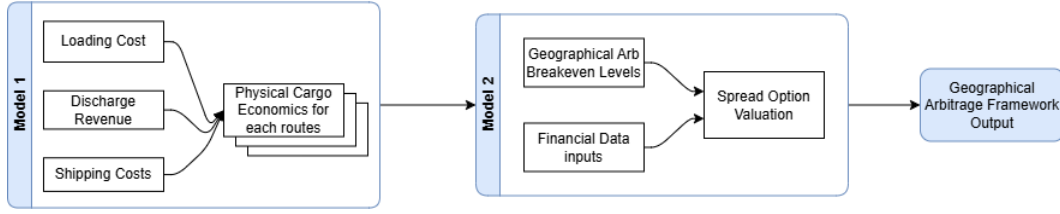


Figure 2.1: Methodologies Overview

2.1 Geographical Arbitrage Analysis

The economic analysis of LNG shipping from export terminals to various continental destinations requires a comprehensive assessment of voyage distances, freight costs, and canal transit fees. Key cost determinants include vessel charter rates, bunker fuel expenses, and terminal fees at receiving ports. Additionally, price differentials between the loading port index benchmark and regional LNG pricing indices (e.g. JKM, TTF) influence netback values and trade flows. Seasonal demand fluctuations, geopolitical risks, and weather-related disruptions further impact the feasibility and cost-effectiveness of specific routes. A comparative cost-per-MMBtu analysis provides a systematic approach to optimizing fleet deployment strategies and identifying the most economically viable shipping routes under prevailing market conditions.

2.1.1 Shipping Routes and Destinations

Defining shipping routes and destinations is the first step in conducting geographical arbitrage analysis. For the ease of elaboration, in this paper we assume loading terminal is in the US Gulf Coast. There are several major continental destinations globally.

Region	Major Destinations
Asia (DES JKTC)	Japan, China, South Korea, India
Europe (DES NWE)	Spain, UK, France, Germany
South America	Brazil, Argentina, Chile
Middle East (DES MEI)	UAE, Kuwait, Egypt

Table 2.1: Major LNG Destinations

The choice of price index in LNG contracts varies by region, with JKM typically used for cargoes destined for Asia, while TTF is commonly applied in Europe and South America. However, there is no strict requirement dictating which index must be used for a particular region. In some cases, for portfolio balancing and risk management purposes, an Asian destination may price contracts using alternative benchmarks such as TTF or Brent, thereby balancing stakeholders' exposure to multi-index rather than one. This flexibility in pricing mechanisms allows market participants to optimize their positions, manage regional price risks, and take advantage of arbitrage opportunities across different LNG markets.

2.1.2 Freight and other Shipping Related Costs

Estimating freight costs involves calculating voyage distance and time, charter rates and fuel costs. Voyage time is usually estimated with vessel speed and voyage distance. Sometimes, routing constraints such as going through canals need to be taken into consideration. Charter rates could be estimated by taking spot and term charter rates published or shared by ship brokers. There are different vessel specs too (e.g. TFDE, MEGI, or Steam Turbine) which will have an impact on boil-off rate and efficiency. Lastly, bunker costs will vary depending on the engine type and bunker prices. Modern vessels have flexibility to change between

high sulfur fuel oil, very low sulfur fuel oil and LNG boil-off gas. Prices of bunkers also differ at different key bunkering hubs.

Additional costs associated with LNG shipping operations include, but are not limited to, port fees, canal transit charges, EU Emission Allowance (EUA) costs, Standby Letter of Credit (SBLC) costs, and other operational expenses. Port costs arise whenever a vessel loads or discharges at liquefaction or regasification terminals. Canal fees are incurred when vessels transit through major waterways, such as the Suez or Panama Canal, with charges varying based on southbound or northbound routes. EUA costs apply exclusively to cargoes destined for Europe, reflecting regulatory requirements for carbon emissions compliance. Additionally, SBLC costs may be imposed when counterparties require bank guarantees to secure transactions, further influencing the overall economics of LNG trade.

These costs are generally relatively fixed across different destinations and counterparties. In a high flat-price environment, cost variations are typically negligible compared to the higher profit margins per cargo. However, in a low flat-price environment, these cost differences become more significant, potentially constituting a substantial portion of the PnL in LNG trading. Traders must account for these costs more carefully when margins are compressed, as they can directly impact the overall profitability of cargo movements and arbitrage strategies.

2.1.3 Geographical Arbitrage Methodology

Once all assumptions and relevant information are compiled, the geographical arbitrage margin can be calculated by comparing the shipping costs of each route alongside the sale price at each destination. In this framework, any parameters related to loading are represented by FOB (Free on Board), while those related to discharging are denoted as DES (Delivered Ex Ship). The PnL for each route is determined using the following equations, and the optimal route for a U.S.-loaded cargo is simply the one that yields the highest PnL among the available options.

$$\text{PnL}_{JKTC} = \text{Price}_{FOB} \times \text{Volume}_{FOB} - \text{Shipping Cost}_{JKTC} - \text{Price}_{DES_JKTC} \times \text{Volume}_{DES_JKTC} \quad (2.1)$$

$$\text{PnL}_{NWE} = \text{Price}_{FOB} \times \text{Volume}_{FOB} - \text{Shipping Cost}_{DES} - \text{Price}_{DES_NWE} \times \text{Volume}_{DES_NWE} \quad (2.2)$$

To effectively manage the hedging of embedded spread optionality within physical contracts, the geographical arbitrage margin is continuously monitored using the following equations. This margin quantifies the economic feasibility of diverting LNG cargoes between regions. A negative margin indicates a financial loss when diverting a cargo from Northwest Europe to the Far East, whereas a positive margin suggests a profitable arbitrage opportunity. By tracking these margins in real time, traders can assess how close the arbitrage opportunity is to becoming feasible and make informed routing decisions accordingly.

$$\text{Arb}_{JKTC_NWE} = (\text{Price}_{DES_JKTC} - \text{Price}_{DES_NWE}) - (\text{Shipping Cost}_{JKTC} - \text{Shipping Cost}_{NWE}) \quad (2.3)$$

The main idea here is to convert the Arb margin to a corresponding JKM/TTF spread level which could be monitored directly in the market. Using the spot and term JKM/TTF spread levels, the Arb margin could be subtracted from the spread levels to obtain break-even JKM/TTF levels for cargoes to go Far East indifferently against Northwest Europe. This break-even level will be used as the strike level in the spread option model in the next section. It is worth noting that, given the different routes, the shipping costs would differ and hence the break-even levels. In general practice, we calculate the shipping costs (\$/MMBtu) in three routes, via Cape of Good Hope, Suez Canal, or Panama Canal.

$$\text{Spread Option Strike} = \text{Price}_{JKM} - \text{Price}_{TTF} - \text{Arb}_{JKTC_NWE} \quad (2.4)$$

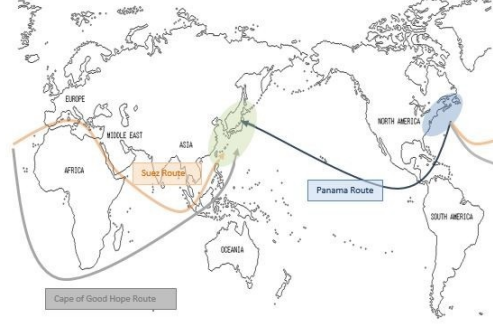


Figure 2.2: Shipping Routes from US to Far East

From the analysis, break-even levels calculated from Panama Canal route is the lowest, i.e. easiest to divert a cargo from Northwest Europe to the Far East. On the other hand, break-even levels calculated from the Cape of Good Hope are the highest. Panama Canal slots are the hardest to book and hence the shipping cost saving might not be achievable in reality. In addition, the canal fee is usually very expensive. It could go as high as two million USD for a round trip. When the freight rate is low, it is not economically viable to sail via the Panama Canal or Suez Canal as the shipping savings are not large enough to recover the canal costs.

Cargo No.	FOB_Date	DES_date	Shipping_Cost_Cape	Shipping_Cost_Panama	Shipping_Cost_Suez
0	09-04-25	23-05-25	1.275545	1.283567	1.246703
1	18-04-25	01-06-25	1.293354	1.288538	1.251530
2	10-05-25	23-06-25	1.328309	1.331774	1.296771
3	17-06-25	31-07-25	1.424977	1.420180	1.388525
4	18-07-25	31-08-25	1.399803	1.390784	1.362671
5	22-08-25	05-10-25	1.423962	1.426845	1.396617
6	11-09-25	25-10-25	1.311544	1.308303	1.278355
7	28-09-25	11-11-25	1.595814	1.603726	1.556751
8	31-10-25	14-12-25	1.856744	1.832632	1.808552
9	28-11-25	11-01-26	1.392558	1.386173	1.356414
10	14-12-25	27-01-26	1.392558	1.386173	1.356414
11	31-12-25	13-02-26	1.370641	1.386173	1.337485

Table 2.2: Shipping Costs across Different Routes

2.2 Spread Option Mathematical Proof

A spread option is an exotic option whose payoff depends on the difference between the prices of two underlying assets. Unlike standard options, which are based on a single asset, the value of a spread option is determined by the spread between the assets' prices at expiration. This proof explores the mathematical models used to value a spread option, focusing on the stochastic processes governing the assets' prices and the methods, such as stochastic calculus, used for pricing and hedging in financial markets.

Overall payoff C_T of Spread Option at expiry is as follows:

$$C(T, S_1(T), S_2(T)) = \max\{S_1(T) - S_2(T) - K, 0\} \quad (2.5)$$

Note: $S_1(T) = JKM$, $S_2(T) = TTF$

Assume C_i follows Geometric Brownian Motion for the prices of underlying assets.

$$\frac{dS_i(T)}{S_i(T)} = \mu_i dt + \sigma_i dw_i(t), \quad i = 1, 2 \quad (2.6)$$

Where: $w_1(t) = \rho \tilde{w}_2(t) + \sqrt{1 - \rho^2} \tilde{w}_1(t)$, $w_2(t) = \tilde{w}_2(t)$

Take expectation and convert the equation to follow Black-Scholes Model to get payoff C_0 at $t = 0$:

$$C_0 = e^{-rT} \mathbb{E}_0 [\max\{S_1(T) - S_2(T) - K, 0\}] \quad (2.7)$$

Solve SDE on individual underlying:

Let $X = \ln S_i(t)$. Using Itô's Lemma, the stochastic differential equation can be written as:

$$dX = \frac{\partial X}{\partial t} dt + \frac{\partial X}{\partial S_i} dS_i + \frac{1}{2} \frac{\partial^2 X}{\partial S_i^2} (dS_i)^2. \quad (2.8)$$

First, calculate the partial derivatives:

$$\begin{aligned} \frac{\partial X}{\partial t} &= 0, \\ \frac{\partial X}{\partial S_i} &= \frac{1}{S_i}, \\ \frac{\partial^2 X}{\partial S_i^2} &= -\frac{1}{S_i^2}. \end{aligned}$$

Now substitute these into the equation for dX :

$$\begin{aligned} dX &= \frac{1}{S_i} dS_i + \frac{1}{2} \left(-\frac{1}{S_i^2} \right) (dS_i)^2, \\ dX &= \frac{1}{S_i} dS_i - \frac{1}{2} \sigma_i^2 dt. \end{aligned}$$

Given the process for $S_i(t)$:

$$dS_i(t) = \mu_i S_i dt + \sigma_i S_i dW_i(t), \quad (2.9)$$

substituting into the equation for dX gives:

$$\begin{aligned} dX &= \frac{1}{S_i} (\mu_i S_i dt + \sigma_i S_i dW_i(t)) - \frac{1}{2} \sigma_i^2 dt, \\ dX &= \mu_i dt + \sigma_i dW_i(t) - \frac{1}{2} \sigma_i^2 dt, \\ dX &= (\mu_i - \frac{1}{2} \sigma_i^2) dt + \sigma_i dW_i(t). \end{aligned}$$

Now, integrating both sides from 0 to T :

$$\begin{aligned}\ln S_i(T) - \ln S_i(0) &= \left(\mu_i - \frac{1}{2}\sigma_i^2\right)T + \sigma_i W_i(T). \\ S_i(T) &= S_i(0) \exp\left(\left(\mu_i - \frac{1}{2}\sigma_i^2\right)T + \sigma_i W_i(T)\right).\end{aligned}\tag{2.10}$$

Next, we would like to apply Change of Numeraire to SDE and reduce two individual stochastic movements into single SDE.

Let $\theta(t)$ be an adapted process satisfying Novikov's condition:

$$\mathbb{E}\left[e^{\frac{1}{2}\int_0^T \theta^2(s)ds}\right] < \infty.$$

Define the Radon–Nikodym process:

$$Z(t) = e^{-\int_0^t \theta(s)dW(s) - \frac{1}{2}\int_0^t \theta^2(s)ds}.\tag{2.11}$$

Then, $Z(t)$ is a martingale and $\mathbb{E}[Z(t)] = 1, \forall 0 \leq t \leq T$. The new Brownian motion is:

$$d\tilde{W}(t) = \theta(t)dt + dW(t).$$

The market price of risk is:

$$\theta(t) = \frac{\alpha(t) - r(t)}{\sigma(t)}.$$

Convert SDE from real world measure to risk neutral measure:

$$\frac{dS_i(T)}{S_i(T)} = r_i dt + \sigma_i d\tilde{W}_i(t), \quad i = 1, 2.$$

Thus,

$$S_i(T) = S_i(0)e^{(r_i - \frac{1}{2}\sigma_i^2)T + \sigma_i \tilde{W}_i(T)}\tag{2.12}$$

Applying Change of Numeraire:

$$S_1^{(S_2)}(T) = \frac{S_1(T)}{S_2(T)}.$$

Using the risk-neutral pricing formula:

$$\mathbb{E}^{(N)}[Y] = \frac{1}{N(0)}\mathbb{E}[YD(T)N(T)],\tag{2.13}$$

where:

$$Y = \max\left(\frac{S_1(T)}{S_2(T)} - 1, 0\right), \quad D(T) = e^{-rT}, \quad N(T) = S_2(T).$$

Thus,

$$C_0 = \mathbb{E}_0\left[e^{-rT} \max(S_1(T) - S_2(T), 0)\right].\tag{2.14}$$

Rewriting:

$$C_0 = e^{-rT} \mathbb{E}_0 \left[S_2(T) \max \left(\frac{S_1(T)}{S_2(T)} - 1, 0 \right) \right] = S_2(0) \mathbb{E}^{(S_2)} \left[\max(S_1^{(S_2)}(T) - 1, 0) \right]. \quad (2.15)$$

Derive $S_1^{(S_2)}(t)$ to retrieve discount rate and volatility needed in the BSM. The dynamics of S_1 and S_2 are:

$$\begin{aligned} \frac{dS_1(T)}{S_1(T)} &= r_f dt + \sigma_1 [\rho d\tilde{W}_2(t) + \sqrt{1 - \rho^2} d\tilde{W}_1(t)], \\ \frac{dS_2(T)}{S_2(T)} &= r_f dt + \sigma_2 d\tilde{W}_2(t). \end{aligned}$$

In vector form:

$$\begin{aligned} \sigma &= (\sigma_1 \rho, \sigma_1 \sqrt{1 - \rho^2}) \\ v &= (\sigma_2, 0) \\ dW &= (d\tilde{W}_1(t), d\tilde{W}_2(t)) \end{aligned}$$

Norm calculation:

$$\begin{aligned} \|\sigma - v\|_2 &= \hat{\sigma} \\ &= \sqrt{\sigma^2 + v^2} \\ &= \sqrt{(\sigma_2 - \sigma_1 \rho)^2 + (\sigma_1 \sqrt{1 - \rho^2})^2} \\ &= \sqrt{(\sigma_1^2 - 2\sigma_1 \sigma_2 \rho + \sigma_2^2)} \end{aligned}$$

Thus,

$$S_1^{(S_2)}(T) = (\rho \sigma_1 - \sigma_2) d\tilde{w}_2' + \sigma_2 \sqrt{1 - \rho^2} d\tilde{w}_1 = \hat{\sigma} dW \quad (2.16)$$

Deriving Stochastic Equations for the Spread by applying the generic equation:

$$\frac{d(X/Y)}{X/Y} = \frac{dX}{X} + \frac{d(1/Y)}{1/Y} + \frac{dX}{X} \frac{d(1/Y)}{1/Y} \quad (2.17)$$

$$\frac{d(1/Y)}{1/Y} = -\frac{dY}{Y} + \frac{dY}{Y} \frac{dY}{Y}, \quad \text{where } X = S_1, Y = S_2 \quad (2.18)$$

Substituting S_1 and S_2 into the equations to get stochastic differential equation for $\frac{dS_1^{(S_2)}(T)}{S_1^{(S_2)}(T)}$:

$$\begin{aligned} \frac{d(1/S_2)}{1/S_2} &= -\frac{dS_2}{S_2} + \frac{dS_2}{S_2} \frac{dS_2}{S_2} \\ &= -r_f dt - \sigma_2 d\tilde{w}_2(t) + \sigma_2^2 dt \end{aligned}$$

$$\frac{dS_1^{(S_2)}(T)}{S_1^{(S_2)}(T)} = \frac{dS_1}{S_1} + \left(-\frac{dS_2}{S_2} + \frac{dS_2}{S_2} \frac{dS_2}{S_2} \right) + \frac{dS_1}{S_1} \left(-\frac{dS_2}{S_2} + \frac{dS_2}{S_2} \frac{dS_2}{S_2} \right)$$

$$\begin{aligned}
&= r_f dt + \sigma_1 [\rho d\tilde{w}_2(t) + \sqrt{1 - \rho^2} d\tilde{w}_1(t)] - r_f dt - \sigma_2 d\tilde{w}_2(t) + \sigma_2^2 dt - \\
&\quad \left(r_f dt + \sigma_1 [\rho d\tilde{w}_2(t) + \sqrt{1 - \rho^2} d\tilde{w}_1(t)] \right) (r_f dt + \sigma_2 d\tilde{w}_2(t) + \sigma_2^2 dt) \\
&= \sigma_1 [\rho d\tilde{w}_2(t) + \sqrt{1 - \rho^2} d\tilde{w}_1(t)] - \sigma_2 d\tilde{w}_2(t) + \sigma_2^2 dt - \sigma_1 \sigma_2 \rho dt \\
&= (\sigma_2 - \sigma_1 \rho) d\tilde{w}_2(t) + \sigma_1 \sqrt{1 - \rho^2} d\tilde{w}_1(t) + \sigma_2 (\sigma_2 - \sigma_1 \rho) dt \\
&= \sigma_1 \sqrt{1 - \rho^2} d\tilde{w}_1(t) + (\sigma_2 - \sigma_1 \rho) (\sigma_2 dt + d\tilde{w}_2(t)) \\
&= \sigma_1 \sqrt{1 - \rho^2} d\tilde{w}_1(t) + (\sigma_2 - \sigma_1 \rho) d\tilde{w}_2^f(t)
\end{aligned}$$

Deriving the Black-Scholes-Merton (BSM) Formula

$$\begin{aligned}
\frac{\partial S_1^{(S_2)}(t)}{\partial t} &= 0 \\
\frac{\partial S_1^{(S_2)}(t)}{S_1^{(S_2)}(t)} &= \frac{1}{S_1^{(S_2)}(t)} \\
\frac{\partial^2 S_1^{(S_2)}(t)}{\partial S_1^{(S_2)}(t)} &= -\frac{1}{(S_1^{(S_2)}(t))^2} \\
d \ln S_1^{(S_2)}(t) &= 0 dt + \frac{dS_1^{(S_2)}(t)}{S_1^{(S_2)}(t)} - \frac{1}{2} \left(\frac{dS_1^{(S_2)}(t)}{S_1^{(S_2)}(t)} \right)^2 \\
d \ln S_1^{(S_2)}(t) &= \sigma_1 \sqrt{1 - \rho^2} d\tilde{w}_1(t) + (\sigma_2 - \sigma_1 \rho) d\tilde{w}_2^f(t) + \frac{1}{2} (\sigma_1^2 + \sigma_2^2 - 2\sigma_1 \sigma_2 \rho) dt \\
S_1^{(S_2)}(T) &= S_1^{(S_2)}(0) e^{\frac{1}{2} (\sigma_1^2 + \sigma_2^2 - 2\sigma_1 \sigma_2 \rho) T + \sigma_1 \sqrt{1 - \rho^2} \tilde{w}_1(T) + (\sigma_2 - \sigma_1 \rho) \tilde{w}_2(T)}
\end{aligned}$$

Applying the Black-Scholes Model

$$\begin{aligned}
C_0 &= S_2(0) E_0^{(S_2)} \left[\max \left\{ S_1^{(S_2)}(T) - 1, 0 \right\} \right] \\
&= S_2(0) \left[\frac{S_1(0)}{S_2(0)} \Phi(d_1) - \Phi(d_2) \right] \\
&= S_1(0) \Phi(d_1) - S_2(0) \Phi(d_2)
\end{aligned} \tag{2.19}$$

where:

$$d_{1,2} = \frac{\ln \left(\frac{S_1(0)}{S_2(0)} \right) \pm \frac{1}{2} \hat{\sigma}^2 T}{\hat{\sigma} \sqrt{T}}$$

Applying Kirk's Approximation:

$$\begin{aligned}
C_T &= \max \{ S_1(T) - S_2(T)', 0 \} \\
S_2(T)' &= S_2(T) + K
\end{aligned}$$

$$\frac{dS_2(T)'}{S_2(T)'} = \frac{d(S_2(T) + K)}{S_2(T) + K} = \frac{dS_2(T)}{S_2(T) + K} = \frac{S_2(T)}{S_2(T) + K}(r_f dt + \sigma_2 d\tilde{w}_2(t)) \quad (2.20)$$

Solve SDE on $S_2(T)'$

$$d \ln S_2'(t) = \left(r' - \frac{1}{2} \sigma_2'^2 \right) dt + \sigma_2' d\tilde{w}_2(t)$$

$$\sigma_2'(t) = \frac{S_2(t)}{S_2(t) + K e^{-r(T-t)}} \sigma_2$$

Apply New Volatility and Discount Rate into BSM

$$C_0 = S_1(0)\Phi(d_1) - (S_2(0) + K e^{-rT})\Phi(d_2) \quad (2.21)$$

where:

$$d_{1,2} = \frac{\ln \left(\frac{S_1(0)}{S_2(0) + K e^{-rT}} \right) \pm \frac{1}{2} \sigma'^2 T}{\sigma' \sqrt{T}}$$

$$\sigma' = \sqrt{[\sigma_2'(0)]^2 + \sigma_1^2 - 2\rho\sigma_2'(0)\sigma_1}$$

2.2.1 Useful Greeks

Delta with respect to corresponding asset price movement.

1. Asset 1:

- Call for Asset 1:

$$\Delta_1^c = \frac{\partial c}{\partial S_1} = e^{-rT} \Phi(d_1) \quad (2.22)$$

- Put for Asset 1:

$$\Delta_1^p = \Delta_1^c - e^{-rT} \quad (2.23)$$

2. Asset 2:

- Call for Asset 1:

$$\Delta_2^c = \frac{\partial c}{\partial S_2} = e^{-rT} \left[-\Phi(d_2) + S_2(T)' \cdot \Phi(d_2) \cdot \sqrt{T} \cdot \frac{\partial \sigma}{\partial S_2} \right] \quad (2.24)$$

where:

$$\frac{\partial \sigma}{\partial S_2} = \frac{\sigma_2 K}{\sigma'} \left(\frac{\sigma_2 S_1(0)}{S_2(0) + K e^{-rT}} - \rho \sigma_1 \right) \Bigg/ \left(\frac{S_1(0)}{S_2(0) + K e^{-rT}} \right)^2$$

- Put for Asset 1:

$$\Delta_2^p = \Delta_2^c + e^{-rT} \quad (2.25)$$

2.3 Dynamic Delta Hedging

Once we have both the necessary information on physical cargoes and their corresponding financial index, we could apply spread option valuation models onto the physical cargo geographical arbitrage framework. Having the option to divert from Europe to Asia is equivalent to holding a call option:

- **Base case:** Loading cargo from the US Gulf Coast and discharge in NWE will make the portfolio short HH and long TTF.
- **Alternative case:** Diverting the cargo from NWE to Asia will make the portfolio long JKM. The transition process is equivalent to having a long position in JKM/TTF. Hence, the option to divert cargo is equivalent to holding a long position in spread call option.

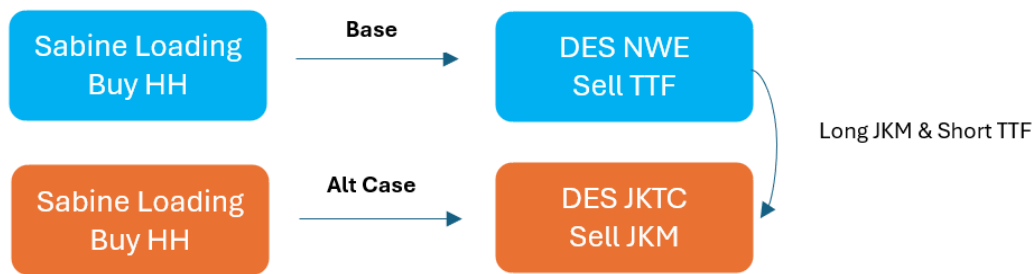


Figure 2.3: Cargo Diversion Framework

By applying equations from section 2.1.3, we could calculate strike for each spread option. Breakeven spread is always negative if arb is closed. Generally under normal market conditions, breakeven spread from route of cape is smaller than route of suez, and subsequently route via panama. In another word, route via cape will give the least economical shipping economics, which is the most out-of-the-money spread option among the three routes.

As the cargo economics getting closer to be in-the-money, we can start delta hedge the spread option by selling JKM/TTF spread in the market. A couple of scenarios could be anticipated:

- 1) The FOB cargo continues to be more in-the-money to be diverted to Far East, and eventually Pnl is high enough to support the physical diversion to Far East from Northwest Europe position.

We continue to delta hedge the spread option through selling JKM/TTF in tranches. When we finally divert the cargo from the West to the East, we fully hedge the position by selling up to 100% of the volume of the cargo size.

- 2) The FOB cargo continues to be more in-the-money to be diverted to Far East, but eventually PnL is NOT high enough to support the physical diversion to Far East from Northwest Europe position.

We continue to delta hedge the spread option through selling JKM/TTF in tranches based on delta of synthetic call spread options. Before final decision-making deadline, we keep the cargo in the West, and we closed the JKM/TTF position by buying back all the previous hedges of the JKM/TTF spread. As the JKM/TTF spread price comes off, we should be overall making money from the “Buy-high-Sell-low” hedging trades.

3 Results

3.1 Spot Cargo Analysis

Taking several spot cargoes loading in 2025 as an example. If we have 10 FOB loading cargoes in the US Gulf Coast, with different loading days spread across April to December, the underlying value of optionality to divert the cargo is calculated with spread option model. Firstly, by calculating purchase costs, sale revenue, shipping costs, and all other associated operational expenses, we could derive the breakeven JKM/TFU levels for each and every cargo. Also, as the loading dates are different, we will have different exercise dates of the decision to divert cargo from West to East. Usually the decision to divert cargo needs to be made around 30 days prior to the delivery window. In this case, if The DES date in Asia is 23rd of May, the decision date must be before 24rd of April. We could see the option value varies with maturity dates and how much out-of-money the option is. All options are worth more than 50c USD, which equivalent to more than 1 million USD of notional option premium for each cargo size of 3.5 Tbtu. The value behind the optionality is tremendous. Such value could be easily overlooked as most people would not expect this kind of options are such valuable. With the output of Delta values, we could further delta hedge each underlying, JKM and TTF, to achieve delta neutral.

	FOB_Date	DES_date	JKM/TFU	Tau(yr)	Spread_option_value	S1_Delta	S2_Delta
0	2025-04-09	2025-05-23	-0.167403	0.236964	0.516290	0.437148	-0.389448
1	2025-04-18	2025-06-01	-0.016100	0.261622	0.615967	0.472496	-0.421666
2	2025-05-10	2025-06-23	-0.016100	0.321896	0.684431	0.474385	-0.418145
3	2025-06-17	2025-07-31	0.059500	0.426005	0.801680	0.479420	-0.414956
4	2025-07-18	2025-08-31	0.009375	0.510937	0.877931	0.481151	-0.410801
5	2025-08-22	2025-10-05	0.070867	0.606827	0.969920	0.485948	-0.409520
6	2025-09-11	2025-10-25	0.070867	0.661622	1.048838	0.498149	-0.418294
7	2025-09-28	2025-11-11	0.320075	0.708197	1.030271	0.480131	-0.398239
8	2025-10-31	2025-12-14	0.609813	0.798608	1.253078	0.520436	-0.433005
9	2025-11-28	2026-01-11	0.501866	0.875320	1.391454	0.545461	-0.454148
10	2025-12-14	2026-01-27	0.501866	0.919156	1.424058	0.545246	-0.451847
11	2025-12-31	2026-02-13	0.416219	0.965731	1.350632	0.524754	-0.429195

Figure 3.1: Spot Cargo Diversion Analysis

3.2 Back testing & Simulation

To further illustrate the benefit of managing delta hedging of the cargoes, I have simulated the outcome of different scenarios in the cases of gradually higher JKM/TFU spread level, gradually lower spread levels and stable JKM/TFU levels. In all three scenarios, the benefit of delta hedging the embedded spread option is to capture the option value and mitigate the risk of extreme scenarios. The outcome of all three scenarios are showing better outcome than cargo pnl itself. In the case of lower JKM/TFU spread levels, delta hedging achieved a pnl of +\$2.1 million USD, which helps to recover some losses from the cargo pnl, which is -\$6.6 million USD. It needs to be highlight that in the case of higher JKM/TFU spread levels, net pnl is reduced

as delta hedging will end up with negative pnl. In the simulated case, the hedge pnl is -\$3.4 million USD, with cargo pnl +\$4.9 million USD.

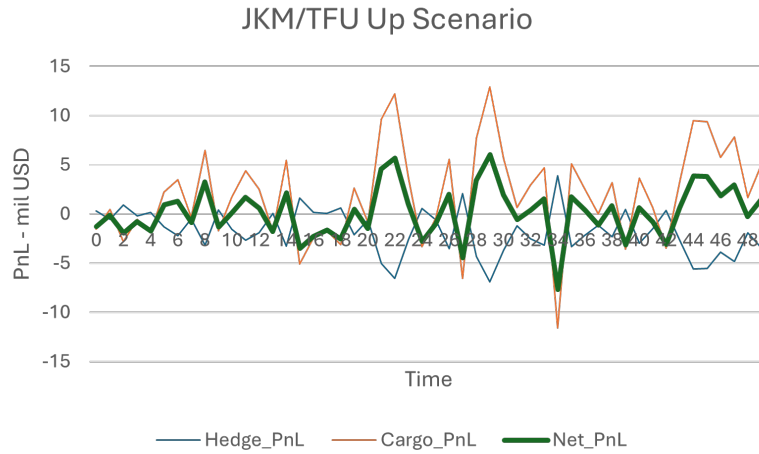


Figure 3.2: JKM/TFU Up Scenario

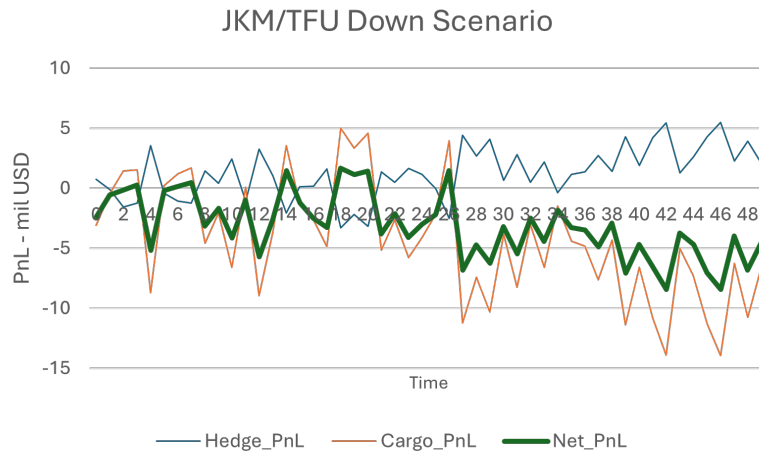


Figure 3.3: JKM/TFU Down Scenario

4 Strength and Weakness

4.1 Strength

4.1.1 Downside Risk Protection

One of the key advantages of employing a spread option hedging strategy is its effectiveness in mitigating downside risk. By dynamically hedging embedded optionality, traders can partially recover losses when market conditions move unfavorably. This ensures greater stability in PnL management and reduces exposure to extreme price swings, which is particularly valuable in the volatile LNG market.

Moreover, this strategy provides a structured approach to risk management by allowing traders to gradually

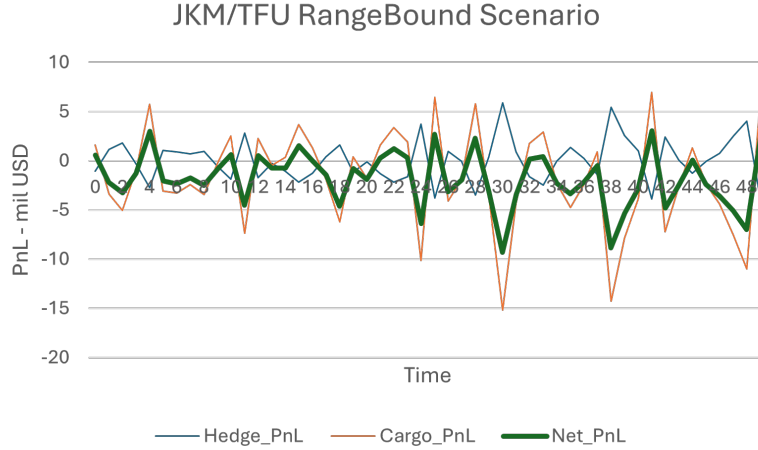


Figure 3.4: JKM/TFU Rangebound Scenario

hedge their positions rather than making abrupt adjustments. This progressive hedging reduces the likelihood of sudden financial shocks, ensuring a smoother PnL trajectory. Additionally, by hedging unfavorable spread movements early, firms can free up capital that might otherwise be tied to margin requirements, allowing for more flexible investment and trading decisions.

4.1.2 Ease of Implementation

Another strength of this strategy is its ease of execution. The required financial instruments, such as JKM and TTF futures, are listed on established exchanges with transparent pricing and transaction history. As long as the contract has not yet entered its pricing window, liquidity is generally sufficient to allow for seamless hedging execution. This accessibility enables market participants to efficiently implement and manage their delta hedging strategies, even with limited operational complexity.

Furthermore, the widespread adoption of algorithmic trading and real-time market data analytics has significantly improved the efficiency of spread option hedging. Many trading platforms offer automated execution tools that can systematically adjust hedging positions in response to market fluctuations. This automation reduces human error, improves response time, and enhances overall trading efficiency. Additionally, regulatory transparency and standardized contract specifications further simplify execution, making the strategy a practical choice for LNG traders of various scales.

4.2 Weaknesses

4.2.1 Capped Profit Potential

One major limitation of managing embedded spread options through delta hedging is the inherent capping of profit potential. By systematically hedging the optionality within physical LNG cargo contracts, traders mitigate downside risk but simultaneously limit potential gains. The profit ceiling is determined by the theoretical value of the spread option, which varies based on hedge frequency and execution levels. As the spread option approaches its expiry, liquidity diminishes, making it increasingly difficult to execute delta hedges effectively. The challenge is particularly pronounced in the JKM market, which remains relatively underdeveloped compared to other major energy trading hubs. Large-scale hedge execution may face slippage, meaning that the execution price deviates unfavorably from the intended level, ultimately

reducing the overall profitability of the strategy. Consequently, traders may forego higher earnings that could have been realized had the position remained fully exposed to market fluctuations.

4.2.2 Misalignment of Expiry Dates Between Financial and Physical Positions

A further complication arises from the misalignment between the expiration of financial derivatives and physical LNG contracts. JKM and TTF futures often expire before the embedded spread option, i.e. when a physical contract reaches its decision point. To mitigate this, traders frequently use second-nearby futures as proxy hedges, introducing basis risk, the spread between first- and second-nearby contracts. While typically manageable, extreme market conditions, such as those seen during the Ukraine crisis, can lead to severe price dislocations, exacerbating financial exposure. A failure to anticipate such risks could lead to substantial losses.

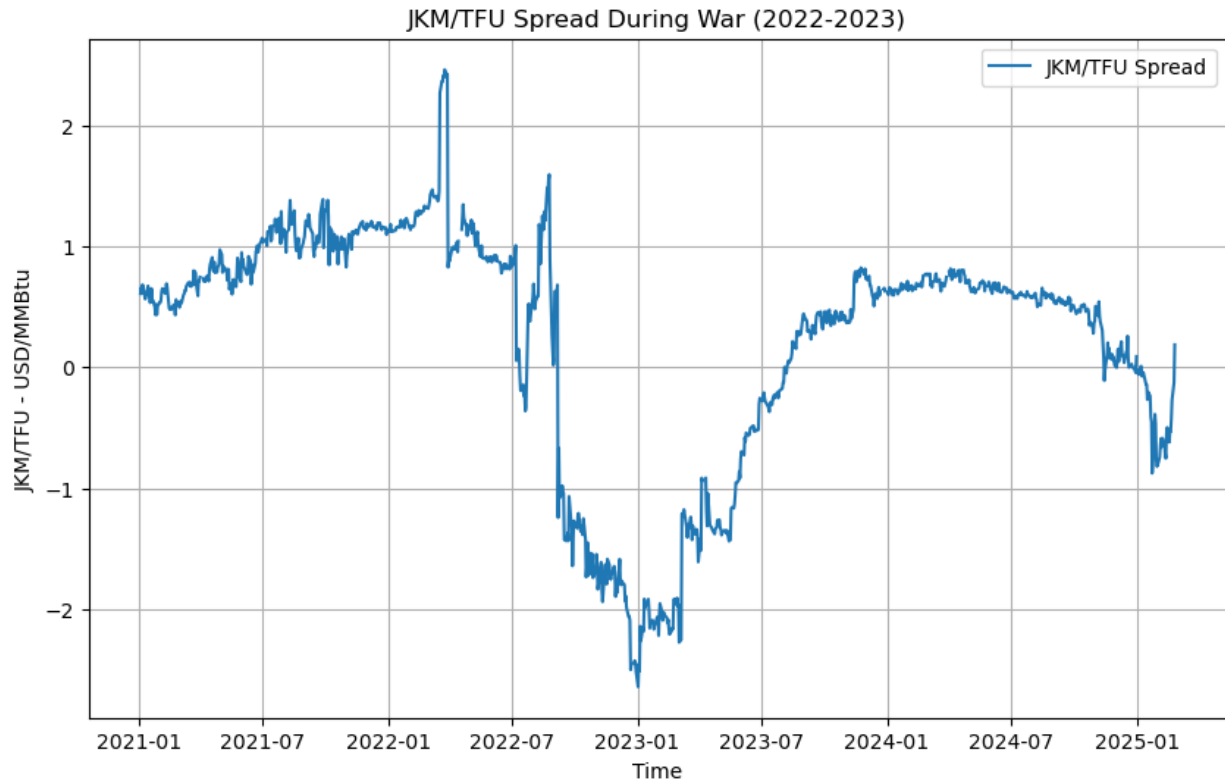


Figure 4.1: JKM/TFU Spread Drastic Movement during Ukraine War

5 Improvements and Future Works

The current framework for managing embedded spread options in LNG trading presents a strong foundation, however, several enhancements could further refine pricing models, back-testing methodologies, and overall trading strategies. Future work should focus on applying spread option technique in long-term LNG contracts, incorporating more advanced statistical models, and exploring alternative risk management techniques.

Spread options are widely used in long-term contracts [12]. They could be used for long-term opportunity pricing. In most long-term contracts, there is a clause called Downward Quantity Tolerance (DQT)

and Upward Quantity Tolerance (UQT), which establish the permissible variations in contractual delivery volumes, providing flexibility for both buyers and sellers. DQT defines the quantity that the buyer could decline to take, within contractually obligation, providing flexibility while ensuring supply commitments, whereas UQT allows the buyer to nominate additional volumes beyond the contracted amount within a predefined limit. For example, in a contract specifying 1 million tonnes per annum (MTPA) with a DQT of 90% and a UQT of 110%, the buyer must receive at least 900,000 tonnes but has the option to procure up to 1.1 MTPA depending on market conditions. These tolerance mechanisms play a critical role in balancing supply and demand uncertainties, supporting market liquidity while safeguarding the commercial interests of both parties. Furthermore, long-term LNG contracts typically uses other indexes such as Brent index or Japanese Crude Cocktail (JCC) index for pricing. DQT and UQT could be perceived as call and put spread options on Brent/JKM or JCC/JKM.

While the study currently employs the Black-Scholes-Merton model and Kirk’s approximation, further refinements can be made by integrating:

- **Stochastic volatility models**, such as the Heston model, to better capture market fluctuations. [13]
- **Machine learning algorithms**, which can provide more dynamic pricing predictions based on historical data patterns.
- **Alternative numerical techniques**, such as Monte Carlo simulations, to increase the accuracy of spread option valuations under different market conditions.
- **Market microstructure considerations**, incorporating real-time trading data to refine bid-ask spreads and transaction costs in pricing calculations.

Lastly, backtesting is crucial for assessing hedging strategies, and enhancements should prioritize incorporating market-reflective inputs to better capture real-world conditions. Key factors such as shipping variations, internal and external discounting rates, and bunker price management must be considered to improve model accuracy and resilience. Even minor discrepancies across different variables can compound into significant financial impacts. Furthermore, integrating real market data ensures continuous calibration of simulations, enabling more precise, adaptive, and effective risk management.

6 Conclusion

This study provides a structured framework for integrating spread option valuation into LNG physical trading, offering a comprehensive approach to managing embedded optionality and geographical arbitrage opportunities. By applying financial derivatives to LNG cargo economics, traders can enhance risk-adjusted decision-making, optimize PnL stability, and systematically monetize arbitrage opportunities. The results indicate that a proactive approach to managing spread options can significantly mitigate financial risks while improving portfolio performance. The study highlights key strengths, such as downside risk protection and ease of implementation, but also acknowledges certain challenges, including profit capping and liquidity constraints.

Future enhancements in pricing models, back-testing methodologies, and contract structuring can further refine the strategy, making it more adaptive to evolving market conditions. As the LNG market continues to grow in complexity, leveraging advanced financial instruments and data analytics will be crucial in maintaining a competitive edge.

Overall, this research lays the foundation for more sophisticated trading strategies and contributes to the broader understanding of how financial derivatives can enhance physical LNG trading. Future studies should

expand on these insights by exploring additional optionality structures and integrating real-time data-driven approaches to improve decision-making in LNG markets.

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